25900912 **A**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Multi-Disciplinary Courses (MDC) - MG1MDCPHY100 - PHYSICS AROUND YOU
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

***[Note: Answers have been given in bold font]**

Part A

Multiple Choice Questions
Each Question carries 1 mark

1. Which of the following is a derived unit? [K] / [CO1]

a) Second b) Meter **c) Pascal** **d) Coulomb**

Give 1 mark for either **c or d**

2. The energy possessed by a body due to its motion is called [K] / [CO1]

a) Potential energy **b) Kinetic energy** c) Elastic energy d) Thermal energy

3. Which of the following is NOT affected by the Doppler Effect? [U] / [CO1]

a) Frequency b) Wavelength **c) Speed of sound in the medium** d) Pitch

4. When a number of forces acting simultaneously on a body bring about a change in its state of rest or of uniform motion in a straight line, then these forces acting on the body are said to be: [K] / [CO1]

a) Balanced forces b) Equal forces **c) Unbalanced forces** d) Opposite forces

5. What must be the angle between the applied force and displacement for doing maximum work? [U] / [CO1]

a) 0 degrees b) 30° c) 45° d) 90°

$W = FdCos\theta$, take angle as zero

6. 1 cubic metre is equal to … cubic centimetre [K] / [CO1]

a) 10^{-6} b) 10^{-4} c) 10^{-2} **d) 10^{6}**

7. The frequency range of sound waves audible to humans is: [K] / [CO1]

a) 20 to 20000 Hz b) 20 to 2000 Hz c) 20 to 20000 kHz d) 20 to 2000 kHz

8. Which of the following is a scalar quantity? [K] / [CO1]

a) Velocity b) Force **c) Speed** d) Acceleration

9. A person standing 1m away from a plane mirror will see his image at a distance of:

[U] / [CO3]

a) 0.5 m **b) 1 m** c) 2 m d) 3 m

10. The refractive index of water is approximately: [K] / [CO3]

a) 0.33 **b) 1.33** c) 1.5 d) 2

11. Which among the following processes is responsible for the working of LASER?

[U] / [CO4]

a) Absorption of photons
b) Stimulated emission
c) Population inversion
d) All of the above

12. Which of the following factors does not affect Resistivity of a conductor? [U] / [CO2]

a) Length b) Area of cross-section c) Material **d) Voltage**

13. By Ohm's Law, how is current and resistance related? [K] / [CO2]

a) Directly proportional **b) Inversely proportional** c) No relation d) Linearly increasing

14. Which of the following affects voltage in an electrical circuit? [U] / [CO2]

a) Resistance b) Current c) Power **d) All of the above**

15. What is the frequency of AC supply in India? [K] / [CO2]

a) 60 Hz **b) 50 Hz** c) 100 Hz d) 230 Hz

Part B
Multiple Choice Questions
Answer **All** Questions
Each Question carries **2** mark

16. A car increases speed from 30 to 40 m/s in 5 s. What is acceleration? [U] / [CO1]

: Hint $v = u + at$

a) 2 m/s² b) 3 m/s² c) 5 m/s² d) 4 m/s²

17. Choose the correct statement [K] / [CO1]

a) distance is a scalar, displacement is a vector, speed is a vector, velocity is a scalar

b) distance is a vector, displacement is a vector, speed is a vector, velocity is a scalar

c) distance is a scalar, displacement is a vector, speed is a scalar, velocity is a vector

d) distance is a scalar, displacement is a vector, speed is a vector, velocity is a vector

18. The speed of a wave is determined by: [U] / [CO1]

a) Wavelength and frequency b) Amplitude and wavelength c) Amplitude and frequency d) Amplitude and period

19. A 1.5 Kg book is held 60 cm above a desk whose top is 70 cm above the floor. Find the Potential energy of the book (a) with respect to the desk (b) with respect to the floor
Hint: Potential Energy=mgh [U] / [CO1]

Correct option: **d) 8.8 J and 19.1 J**

Solution:
At height 0.6 m → PE = 1.5 × 9.8 × 0.6 ≈ 8.8 J
At height 1.3 m → PE ≈ 19.1 J

20. Which of the following is NOT a unit of Power? [U] / [CO1]

a) Watt b) Horsepower **c) Newton metre** d) Newton metre/second

21. If the angle of incidence exceeds the critical angle, what occurs? [A] / [CO3]

a) Refraction b) Absorption **c) Total internal reflection** d) Scattering

22. What is the power in a circuit with 5V voltage and 2A current? [U] / [CO2]

Power= VI = 10 W (5*2)
Answer: a) 10 W

23. Which of the following does not affect Rayleigh scattering? [A] / [CO3]

a) Wavelength b) Particle size c) Intensity **d) Temperature of the sun**

24. How is the energy of an electromagnetic wave related to its frequency? [U] / [CO3]

a) Directly proportional b) Inversely proportional c) Exponential d) Not related

25. For what purpose is Laser is used in LIDAR ? [U] / [CO4]

a) High speed photography

b) Optical carrier signal

c) Drilling

d) Range finder